

ATPL EASA STUDY GUIDE | 032 - PERFORMANCE

Chapter 5: Class A Take-Off Speeds (CS-25 Hierarchy: V1, Vr, V2)

Syllabus Source: CAE Oxford (pp. 149-194) | Pass Target: >= 90% AviationExam

1. THE CS-25 TAKE-OFF SPEEDS SPECTRUM

Class A covers commercial jet transports certified under CS-25. Take-off is governed by an interlocking hierarchy of calibrated airspeeds:

Speed Code & Name	Aerodynamic & Regulatory Definition	Mandatory Legal Constraint / Relationship
Speed Code & Name	Aerodynamic & Regulatory Definition	Mandatory Legal Constraint / Relationship
Vs1g / Vsr (Reference Stall)	Stall speed at 1g normal acceleration in take-off configuration.	Baseline reference for all minimum safety speeds.
Vmcg (Min Control Ground)	Minimum speed to maintain directional control on runway using RUDDER ONLY following engine failure.	Vmcg <= V1! (If engine fails below Vmcg, rudder cannot keep plane on runway!).
V1 (Decision Speed)	Maximum speed to abort take-off; minimum speed to continue take-off safely following engine failure.	Vmcg <= V1 <= Vr and V1 <= V_MBE (Max Brake Energy Speed).
Vr (Rotation Speed)	Speed at which pilot initiates backward pull on control column to pitch up.	Vr >= V1 and Vr >= 1.05 x Vmca.
Vmu (Min Unstick Speed)	Minimum speed at which aircraft can safely lift off ground without tail strike.	Demonstrated during flight test with tail dragging on runway.
Vlof (Lift-Off Speed)	Actual speed at which the main tires leave the runway surface.	Vlof >= 1.10 x Vmu (all engines) or >= 1.05 x Vmu (OEI).
V2 (Take-off Safety Speed)	Target climb speed reached at 35 ft screen height with one engine inoperative.	V2 >= 1.13 x Vsr and V2 >= 1.10 x Vmca!

2. THE GOLDEN HIERARCHY OF TAKE-OFF SPEEDS

The strict sequential order of speeds during an accelerating take-off roll is:

- The Take-off Sequence: $V_{mcg} \leq V1 \leq Vr < V_{lof} < V2$. (Never can V1 be less than Vmcg, and never can V1 exceed Vr!).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Class A Jet Takeoff Speed Chain

Master Speed Chain: $V_{MCG} \leq V_{EF} < V1 \leq Vr \leq V_{LOF} < V2$. - V_{MCG} : Minimum control speed on ground using aerodynamic rudder only. - $V1$: Takeoff decision speed. - Vr : Rotation speed ($V1 \leq Vr$). - $V2$: Takeoff safety speed at 35 ft screen height ($V2 \geq 1.13 VSR$). $V1$ Action Rule: 1. Prior to $V1$: ABORT TAKEOFF immediately for engine failure, fire, or major system failure. 2. At or above $V1$: CONTINUE TAKEOFF into the air, even with total engine failure! Stopping past $V1$ leads to catastrophic runway excursion.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Checking Take-Off Speeds Legality

SCENARIO (Classic EASA Flight Deck Trap Drill): A dispatch calculation for a Boeing 737 yields the following calculated speeds: - $V_{mcg} = 112 \text{ kt}$ - $V_{mca} = 118 \text{ kt}$ - $V_1 = 110 \text{ kt}$ - $V_r = 122 \text{ kt}$ - $V_2 = 130 \text{ kt}$ QUESTION: Is this speed schedule legal for departure under CS-25 regulations? If not, what must be corrected? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check Constraint 1: $V_{mcg} \leq V_1$: - $V_{mcg} = 112 \text{ kt}$ - $V_1 = 110 \text{ kt}$ - Check: Is $112 \leq 110$? NO! V_1 (110 kt) is LESS than V_{mcg} (112 kt)! - WHY THIS IS DANGEROUS FOR DUMMIES: If an engine fails at 111 kt, you are above V_1 (committing to fly), but below V_{mcg} (meaning the rudder cannot keep the aircraft straight on the runway, causing a runway excursion!). - Mandatory Fix: V_1 MUST be raised to at least 112 kt ($V_1 \geq V_{mcg}$)! Step 2: Check Constraint 2: $V_r \geq 1.05 \times V_{mca}$: - Minimum allowable $V_r = 1.05 \times 118 \text{ kt} = 123.9 \text{ kt}$ (~124 kt). - Actual proposed $V_r = 122 \text{ kt}$. - Check: Is $122 \geq 124$? NO! V_r is 2 knots too low! - Mandatory Fix: V_r must be raised to at least 124 kt! FINAL ANSWER: The schedule is ILLEGAL on two counts! V_1 must be increased to at least 112 kt, and V_r must be increased to at least 124 kt.

[!] EXAM TRAP (AVIATIONEXAM) - The V_1 Decision Point Trap

- V_1 is the DECISION SPEED: The action to abort must be INITIATED BY V_1 ! - In modern regulations, V_1 is the MAXIMUM speed by which the pilot must have made the decision AND applied the first stopping action (brakes/spoilers). If an engine fails after V_1 , YOU MUST CONTINUE THE TAKE-OFF!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Under CS-25 regulations, what is the strict relationship between V_1 (Decision Speed) and V_{mcg} (Minimum**

- [A]: V_1 must be less than V_{mcg}
- [B]: V_1 must be GREATER THAN OR EQUAL TO V_{mcg} ($V_1 \geq V_{mcg}$)
- [C]: V_1 is always 1.2 times V_{mcg}
- [D]: V_{mcg} only applies in flight

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If V_1 were below V_{mcg} , an engine failure between V_1 and V_{mcg} would require continuing the take-off, yet the aerodynamic rudder force would be insufficient to prevent the aircraft veering off the runway.

Q2: Under CS-25, what is the minimum legal ratio for Take-Off Safety Speed (V_2) relative to reference stall speed

- [A]: $V_2 \geq 1.13 \times V_{sr}$
- [B]: $V_2 \geq 1.05 \times V_{sr}$
- [C]: $V_2 = V_{sr}$
- [D]: $V_2 \geq 1.40 \times V_{sr}$

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. Under CS-25.107, V_2 must provide a stall margin of at least 13% above reference stall speed: $V_2 \geq 1.13 V_{sr}$ (and at least 1.10 V_{mca}).

Q3: If an engine fails at a speed HIGHER than V_1 on a commercial airliner take-off, what is the legally mandated

- [A]: Immediately apply maximum wheel braking and reverse thrust
- [B]: CONTINUE the take-off, rotate at V_r , and climb away at V_2

- [C]: Extend full flaps
- [D]: Cut the remaining engine

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Above V1, remaining runway is insufficient to stop the aircraft without overrunning. The aircraft is certified to safely climb away on the remaining engine(s).